

Author

- Laurent d'Orazio, Univ Rennes, CNRS, IRISA

1 Introduction

The application to be considered relies on a collection of games, in particular focusing on a set companies, games and platforms. Figures 1a, 1b and 1c present the data schemas.

Company
id:int
guid:string
name:string
abbreviation:string
creation:string
city:string
state:string
country:string
image:string

(a) Data schema
Company

Game
id:int
guid:string
title:string
release:string
cover:string

(b) Data schema
Game

Platform
id:int
guid:string
name:string
abbreviation:string
release:string
price:string
company:int
image:string

(c) Data schema
Platform

Figure 1: Data schemas overview

2 Deployment

Docker pull command

```
docker pull mongo
```

Server (first time)

```
docker run -d -p 27017-27019:27017-27019 --name my_mongo mongo
```

Server (if already exists)

```
docker start my_mongo
```

Sending data file

```
docker cp games.json my_mongo:/tmp/games.json
docker cp companies.json my_mongo:/tmp/companies.json
docker cp platforms.json my_mongo:/tmp/platforms.json
```

Client shell

```
docker exec -it my_mongo bash
mongosh
```

or

```
docker exec -it my_mongo mongosh
```

Importing data

In the OS prompt (not in Mongo client shell):

```
mongoimport --db games --collection game --type json --file /tmp/games.json --jsonArray
mongoimport --db games --collection company --type json --file /tmp/companies.json --jsonArray
mongoimport --db games --collection platform --type json --file /tmp/platforms.json --jsonArray
```

Note: The following queries can be written on string values (for CSV with). However, it is possible to convert fields in their appropriate types using the following command in Mongo client shell:

```
db.platform.updateMany({}, [
  {$set:{
    id:{$toInt: "$id"},
    release_temp:{$toDate: "$release_temp"},
    price:{$toDouble: "$price"}
  } }
])
```

3 Tutorial

The following example enables to retrieve the number of companies par country.

3.1 Map

```
var map=function(){emit(this.country,1) }
```

3.2 Reduce

```
var reduce = function(country,nb){return Array.sum(nb);} 
```

3.3 Execution

```
db.company.mapReduce(map,reduce,{out : "nbCountry"});
db.nbCountry.find()
```

4 Activities

1. Number of companies per city.
2. Names of the companies.
3. Companies where country is France.
4. Most expensive platform per company.
5. Oldest company.
6. Country with the maximum number of companies.

5 References

- **Lecture:** https://people.irisa.fr/Laurent.Dorazio/data/teachings/big_data/big_data_mapreduce.pdf